# Class 7 Social Science – Chapter 3: Climates of India

NCERT – Exploring Society: India and Beyond | Part 1 | Reprint 2026–27

इस अध्याय Climates of India के सभी मुख्य टॉपिक्स, पाठ्यपुस्तक के प्रश्नों एवं गतिविधियों के उत्तर और परीक्षा में पूछे जा सकने वाले महत्वपूर्ण प्रश्न-उत्तर नीचे simple and easy English में दिए गए हैं। उत्तर कक्षा 7 के विद्यार्थियों के स्तर के अनुसार रखे गए हैं।

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# Topic 1: Weather, Seasons and Climate

Q1. What is weather?

Answer: Weather is the condition of the atmosphere at a particular place and time. It may be hot, cold, rainy, windy or dry.

Q2. What is climate?

Answer: Climate is the pattern of weather that a place or region experiences over a long period of time, usually several decades.

Q3. What is the difference between weather and climate?

Answer: Weather changes from hour to hour or day to day. Climate describes the usual pattern of weather over a long period.

Q4. What are seasons?

Answer: Seasons are periods of the year that occur in a regular cycle. Each season lasts for a few months and returns every year.

Q5. How are seasons related to weather and climate?

Answer: Weather changes with the seasons. The regular pattern of seasons is also closely connected with the climate of a region.

Q6. Name the four main seasons found in most parts of the world.

Answer: The four main seasons are spring, summer, autumn and winter.

Q7. Which additional season is important in India?

Answer: The rainy season, also called the monsoon season, is especially important in India.

Q8. How many traditional seasons, or ritus, are recognised in many parts of India?

Answer: Six traditional seasons are recognised.

Q9. Name the six traditional Indian seasons.

Answer:

1. Vasanta – Spring
2. Grishma – Summer
3. Varsha – Rainy season
4. Sharad – Autumn
5. Hemanta – Pre-winter
6. Shishir – Winter

Q10. Name two festivals associated with the traditional seasons.

Answer: Vasanta Panchami and Sharad Purnima are associated with traditional seasons.

Q11. How do seasons affect plants and animals?

Answer: Some plants flower in spring, some trees shed their leaves in autumn, and some animals grow thicker fur during winter.

Q12. How do seasons affect human life?

Answer: Seasons affect the clothes we wear, the food we eat, the crops we grow and the activities we perform.

Q13. Does climate always remain the same?

Answer: Climate usually remains stable over long periods, but scientists have recorded significant changes in the world's climate in recent decades.

Q14. What is the main difference between a season and climate?

Answer: A season is a recurring period within a year. Climate is the long-term pattern of weather in a region.

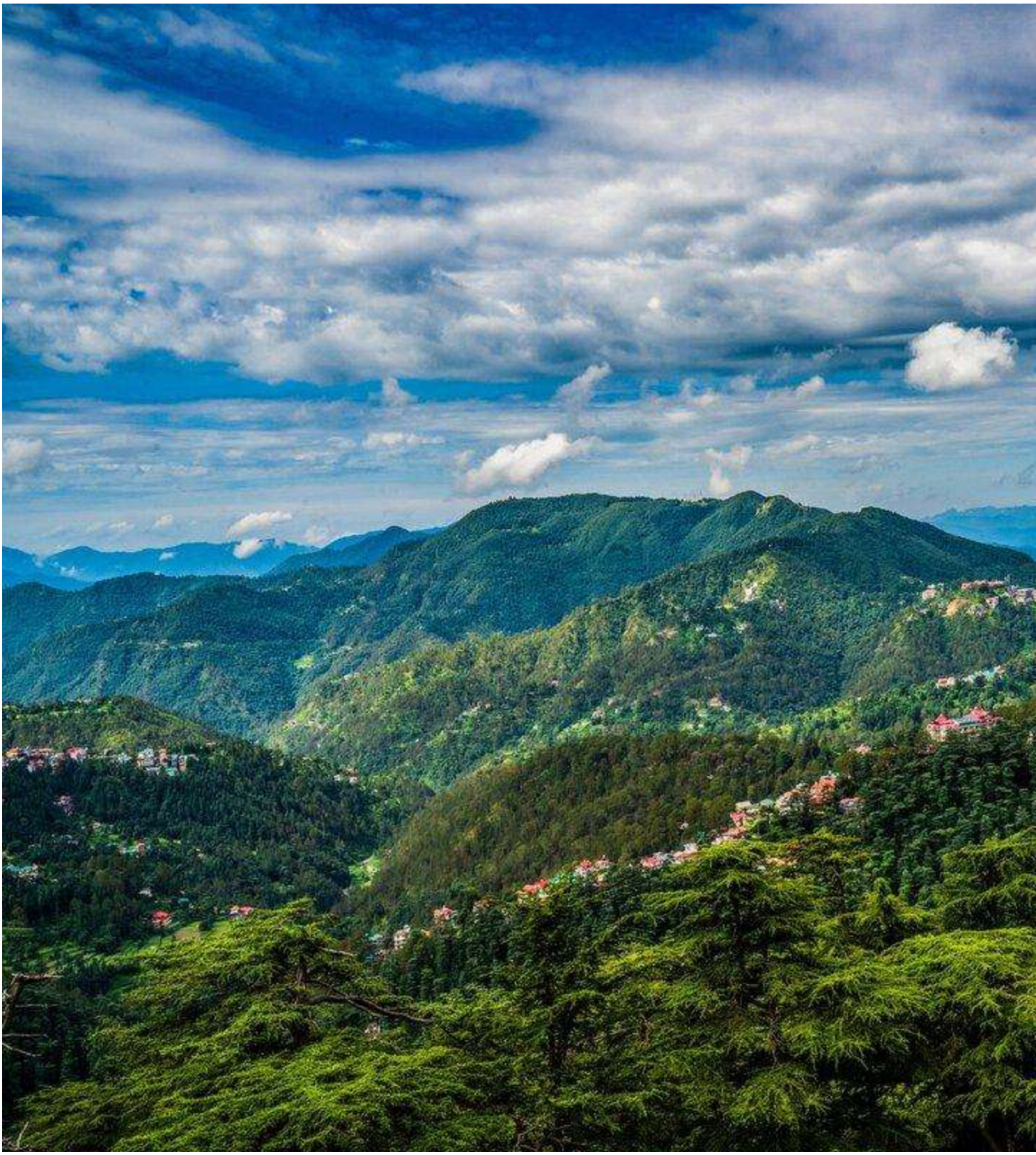
# Topic 2: Types of Climates in India

India has a great variety of climates because of its mountains, plains, deserts, plateaus and long coastline.



1. Alpine Climate

Found in the high Himalayan mountains. Winters are cold and snowy, while summers are cool.



2. Temperate Climate

Found in the lower Himalayas and many hilly areas. Winters are moderately cold and summers are not too hot.



3. Subtropical Climate

Found in the northern plains. Summers are very hot and winters are cold. Much of India's wheat is grown here.



4. Arid Climate

Found in the Thar Desert in western India. Days are extremely hot, nights are cooler and rainfall is very low.



5. Tropical Wet Climate

Found along the western coastal strip. It receives heavy monsoon rainfall and supports rice and spice cultivation.



6. Semi-arid Climate

Found in the central Deccan Plateau. Summers are hot, winters are mild and rainfall is moderate during the rainy season.



7. Tropical Climate

Found in eastern India and the southern peninsula. Winters are mild, with distinct wet and dry periods controlled by monsoon winds.

Q1. Why does India have different types of climates?

Answer: India has different climates because its regions differ in latitude, altitude, distance from the sea, winds and physical features.

Q2. What is an alpine climate?

Answer: It is the climate of high mountain regions, with cold, snowy winters and cool summers.

Q3. What is a temperate climate?

Answer: It is a climate with moderately cold winters and summers that are not too hot.

Q4. What is a subtropical climate?

Answer: It is the climate of the northern plains, with very hot summers and cold winters.

Q5. What is an arid climate?

Answer: An arid climate is a very dry climate with little rainfall.

Q6. What is a tropical wet climate?

Answer: It is a climate with heavy rainfall, especially during the monsoon months.

Q7. What is a semi-arid climate?

Answer: It is a relatively dry climate with limited or moderate rainfall.

Q8. Why is the western coastal strip suitable for rice and spices?

Answer: It receives heavy monsoon rainfall, which supports the growth of rice and many spices.

Q9. Why are hill stations popular during summer?

Answer: Hill stations are at higher altitudes and usually have cooler temperatures than the plains.

Q10. Which climate is found in the Thar Desert?

Answer: The Thar Desert has an arid climate.

# Topic 3: Factors Determining the Climate of India

The textbook explains five main factors that determine the climate of a region.

## 3.1 Latitude

Q1. What is latitude?

Answer: Latitude measures the distance of a place north or south of the Equator.

Q2. How does latitude affect climate?

Answer: Places near the Equator are generally warmer because the Sun's rays fall more directly on them. Places near the poles are colder because the rays spread over a larger area.

Q3. Why are the polar regions colder than the equatorial regions?

Answer: The Sun's rays reach the polar regions at a slant and spread over a larger area. They also pass through more of the atmosphere, so less heat reaches the ground.

Q4. Why is Kanniyakumari generally warmer than Srinagar?

Answer: Kanniyakumari is closer to the Equator, while Srinagar is farther north. Therefore, Kanniyakumari is generally warmer.

## 3.2 Altitude

Q1. What is altitude?

Answer: Altitude is the height of a place above sea level.

Q2. How does altitude affect temperature?

Answer: Temperature generally decreases as altitude increases.

Q3. Why are hill stations cooler than the plains?

Answer: Hill stations are located at higher altitudes, where the air is less dense and temperatures are generally lower.

Q4. Why are many Himalayan peaks covered with snow?

Answer: Many Himalayan peaks are so high that their temperatures remain below the freezing point of water.

Q5. Name four hill stations mentioned in the chapter.

Answer: Shimla, Nainital, Darjeeling and Ooty.

Q6. Ooty and Coimbatore are at almost the same latitude. Why is Ooty cooler?

Answer: Ooty is at a much higher altitude than Coimbatore. Temperature decreases with increasing altitude, so Ooty is cooler.

# 3.3 Proximity to the Sea

Q1. What does proximity to the sea mean?

Answer: It means how near or far a place is from the sea.

Q2. How does the sea affect the temperature of coastal areas?

Answer: The sea heats up and cools down more slowly than land. It reduces extreme temperatures, making coastal summers cooler and winters milder.

Q3. Why do coastal areas have a moderate climate?

Answer: The sea acts as a moderator of temperature because water heats up and loses heat more slowly than land.

Q4. Why does Mumbai have a smaller temperature range than Nagpur?

Answer: Mumbai is near the sea, which moderates its temperature. Nagpur is farther inland and experiences hotter summers and colder winters.

Q5. What temperature range does the textbook give for Mumbai and Nagpur?

Answer: Mumbai has a range of about 14°C, while Nagpur has a range of about 34°C.

Q6. What is the difference between the temperature range of Nagpur and Mumbai?

Answer: \(34-14=20^\circ C\). Nagpur's temperature range is about 20°C greater than Mumbai's.

# 3.4 Winds

Q1. How do winds affect climate?

Answer: Winds carry warm or cool air and can change temperature, humidity and rainfall.

Q2. How do winds from deserts affect northern and central India in summer?

Answer: Dry, hot winds can cause severe heat waves in states such as Punjab, Haryana, Rajasthan and Madhya Pradesh.

Q3. How do winds from the sea affect the land?

Answer: Winds from the sea carry moisture over the land and may bring rainfall.

Q4. What causes cold waves in the Himalayan foothills?

Answer: Cold winds coming from across the Himalayas can cause cold waves.

Q5. How are winds connected with precipitation?

Answer: Winds carry moisture. When moist air cools and water vapour condenses, precipitation may occur.

## 3.5 Topography

Q1. What is topography?

Answer: Topography means the physical features of an area, such as mountains, hills, valleys, slopes and coasts.

Q2. How do the Himalayas and Karakoram ranges affect India's climate?

Answer: They protect much of the Indian subcontinent from the cold winds of the deserts of Central Asia.

Q3. Why does the Thar Desert experience hot and dry winds?

Answer: Its nearly flat topography offers little protection from hot and dry winds.

Q4. How does topography affect rainfall?

Answer: Mountains can act as barriers to moist winds. Some slopes receive heavy rainfall, while areas on the other side may receive much less.

## 3.6 Microclimate

Q1. What is a microclimate?

Answer: A microclimate is the climate of a small area that differs from the climate of the surrounding region.

Q2. Give two examples of microclimates.

Answer: Enclosed valleys and urban heat islands are examples of microclimates.

Q3. What is an urban heat island?

Answer: An urban heat island is a city or urban area that is warmer than its surroundings because buildings and concrete surfaces trap heat and there is often little vegetation.

Q4. How do microclimates affect living things?

Answer: They can affect local plants, animals, crops, human health and well-being.

Q5. How long should climate patterns generally be studied to describe a region's climate?

Answer: The textbook refers to a period of three decades or more.

## Quick Revision Table

| Factor | Main effect on climate |
|---|---|
| Latitude | Creates differences in climate between northern and southern regions. |
| Altitude | Higher places are generally cooler. |
| Proximity to the sea | Moderates temperature. |
| Winds | Carry warm or cool air and affect moisture and rainfall. |
| Topography | Mountains, valleys and other landforms influence temperature and rainfall. |

# Topic 4: The Monsoons






MAWSYNRAM
VILLAGE

8

Q1. What is a monsoon?

Answer: A monsoon refers to seasonal winds that blow over a large region and change direction with the seasons. In everyday use, the word is often associated with seasonal rains.

Q2. From which language does the word monsoon come?

Answer: It comes from the Arabic word mausim, meaning “season”.

Q3. Why are monsoons important for India?

Answer: Monsoons fill rivers, provide water for crops, support agriculture and sustain many livelihoods.

Q4. What causes the monsoon winds?

Answer: Land heats up and cools down faster than the ocean. These differences create pressure differences that cause winds to change direction with the seasons.

Q5. How does the summer monsoon form?

Answer: In summer, the Asian landmass heats up and develops low pressure. Cooler air from the relatively high-pressure ocean moves towards the land, carrying moisture that produces rainfall.

Q6. In which direction do the summer monsoon winds blow?

Answer: The southwest monsoon winds come from the southwest towards India.

Q7. When do the monsoon rains typically begin to advance into India?

Answer: They typically begin advancing from the southern tip of India in early June.

Q8. When do the monsoon rains generally cover the entire Indian subcontinent?

Answer: They generally cover the entire subcontinent by mid-July.

Q9. What is the role of the Western Ghats in the summer monsoon?

Answer: The Western Ghats act as a natural barrier. Their western slopes receive heavy rainfall, while the Deccan Plateau to the east receives less rainfall, often with interruptions.

Q10. What happens to the monsoon winds in winter?

Answer: The land cools faster than the ocean. The winds reverse and generally blow from the land towards the ocean.

Q11. What is the northeast monsoon?

Answer: The northeast monsoon is the winter monsoon. Some of its winds pass over the Bay of Bengal, collect moisture and bring rainfall to parts of eastern and southern India.

Q12. Why do the winter monsoon winds bring rainfall to parts of eastern and southern India?

Answer: Some winds pass over the Bay of Bengal, collect moisture and carry it towards these regions.

Q13. Which place in Meghalaya receives very high average annual rainfall according to the chapter?

Answer: Mawsynram receives about 11,000 mm of average annual rainfall, or about 11 metres, according to the textbook.

Q14. How have monsoons influenced Indian classical music?

Answer: Monsoons have inspired ragas in both Carnatic and Hindustani classical music. Meghamalhar and Amruthavarshini are two examples mentioned in the chapter.

Q15. How did traditional communities predict the arrival of the monsoon?

Answer: They observed natural signs. For example, fishermen on the Konkan coast watched for certain fish appearing at the surface, and some communities observed the flowering of the Golden Shower tree.

Q16. What is the traditional belief about crows building nests high or low in trees?

Answer: Some communities believe that high nests indicate less rainfall, while lower nests indicate that heavy rainfall is likely. This is a traditional belief, not a guaranteed scientific prediction.

# Summer and Winter Monsoon: Difference

| Summer / Southwest Monsoon | Winter / Northeast Monsoon |
|---|---|
| Land becomes hotter than the ocean. | Land cools faster than the ocean. |
| Low pressure develops over the land. | Relatively high pressure develops over the land. |
| Winds blow from the ocean | Winds generally blow from the land towards the ocean. |

towards the land.

| | |
|---|---|
| Moist winds bring widespread rainfall. | Winds are generally dry, but some collect moisture over the Bay of Bengal and bring rain to parts of eastern and southern India. |

# Topic 5: Climate and Our Lives

## 5.1 Climate, Culture and Festivals

Q1. How does climate influence Indian culture?

Answer: Climate influences festivals, food, clothing, dances, songs and agricultural traditions.

Q2. Why are many Indian festivals connected with seasons?

Answer: Many festivals celebrate sowing, harvesting, seasonal changes and the availability of crops.

Q3. Name six Indian festivals connected with seasons or agriculture shown in the chapter.

Answer: Bihu, Baisakhi, Pongal, Onam, Makar Sankranti and Chhath Puja.

Q4. How does the rainy season affect people's lives?

Answer: It provides water for crops and rivers, but excessive or insufficient rainfall can also cause floods, water shortages and agricultural problems.

Q5. How does climate influence the food we eat?

Answer: The crops that grow in a region depend partly on its temperature and rainfall. Therefore, local food habits are often connected with climate.

## 5.2 Climate and the Economy

Q1. What is meant by monsoon failure?

Answer: Monsoon failure refers to insufficient or poor rainfall during the monsoon season.

Q2. How does monsoon failure affect agriculture?

Answer: It reduces the availability of water for crops and may lead to crop failure and lower agricultural production.

Q3. How can poor monsoon rainfall affect food prices?

Answer: Poor rainfall may reduce the production of grains, vegetables and fruits. Their prices may rise, causing inflation.

Q4. How can monsoon failure affect rural workers?

Answer: Agricultural labourers may lose work and may have to migrate to cities in search of employment.

Q5. How can climate affect industrial activity?

Answer: Industries often depend on predictable weather and a reliable supply of water. Changes in climate can disturb their activities.

Q6. How can excessive rainfall affect the economy?

Answer: Excessive rainfall can damage crops, roads, bridges, houses and other infrastructure, causing financial losses.

Q7. Why is a predictable monsoon important for India?

Answer: It supports farming, water availability, food production and the livelihoods of many people.

# Topic 6: Climates and Disasters

The textbook discusses four important climate-related disasters: cyclones, floods, landslides and forest fires.

## 6.1 Cyclones

Q1. What is a cyclone?

Answer: A cyclone is a powerful rotating weather system associated with low atmospheric pressure, strong winds and often heavy rainfall.

Q2. How are cyclones formed?

Answer: A low-pressure system develops over the sea. Air from surrounding areas moves towards it, carrying moisture. When the system becomes intense and the winds become very strong, a cyclone may form.

Q3. What is the eye of a cyclone?

Answer: The eye is the central region of a cyclone. The textbook describes it as a cloudless centre towards which the surrounding winds rotate.

Q4. Which part of India's coastline frequently experiences cyclones?

Answer: The eastern coast of India frequently experiences cyclones.

Q5. What damage can cyclones cause?

Answer: Cyclones can cause loss of human and animal life, damage buildings and infrastructure, uproot trees and cause soil erosion.

Q6. Which organisation tracks cyclones in India?

Answer: The India Meteorological Department (IMD) tracks cyclones and provides information about their development and expected landfall.

Q7. What is the National Disaster Response Force (NDRF)?

Answer: The NDRF is a specially trained force that responds to natural and human-made disasters. It helps with rescue and evacuation during cyclones, floods and landslides.

# 6.2 Floods

Q1. What is a flood?

Answer: A flood occurs when water overflows onto land that is normally dry.

Q2. What are the main causes of floods?

Answer: Floods can be caused by heavy rainfall, excessive runoff, overflowing rivers and lakes, blocked drainage systems and glacial lake bursts.

Q3. Which Indian states are particularly vulnerable to floods according to the chapter?

Answer: Uttar Pradesh, Bihar, Kerala, Andhra Pradesh and Assam are particularly vulnerable.

Q4. Why do floods occur frequently during the monsoon?

Answer: Heavy monsoon rainfall can produce more runoff than the land and drainage systems can handle. Rivers may overflow their banks.

Q5. What is a glacial lake burst?

Answer: It occurs when water from a glacial lake breaks through the barrier of rocks and ice holding it back.

Q6. What happened in Uttarakhand in 2013, according to the textbook?

Answer: Continuous heavy rain caused a sudden glacial burst and many landslides. Areas around Kedarnath were severely damaged, villages were washed away and roads and bridges were destroyed. The textbook reports that about 6,000 people lost their lives.

Q7. Why do cities sometimes experience floods even after heavy rainfall for a short period?

Answer: Poor drainage, construction that blocks waterways and concrete surfaces that prevent water from soaking into the ground can cause urban flooding.

Q8. How can floods be reduced?

Answer: Better drainage, protection of waterways, proper urban planning, flood warnings and avoiding construction in flood-prone areas can help reduce flood risks.

# 6.3 Landslides

Q1. What is a landslide?

Answer: A landslide is the sudden collapse or movement of rock, soil or debris down a slope.

Q2. What are the main causes of landslides?

Answer: Heavy rain, earthquakes and volcanic activity can trigger landslides.

Q3. Name some regions of India where landslides are common.

Answer: Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh and the Western Ghats.

Q4. How do human activities increase the risk of landslides?

Answer: Cutting down forests, constructing buildings without proper methods and blocking the natural flow of water can increase landslide risk.

Q5. How can landslides be prevented or reduced?

Answer: Protecting forests, improving drainage, using safe construction methods and avoiding unstable slopes can reduce the risk.

# 6.4 Forest Fires

Q1. What is a forest fire?

Answer: A forest fire is an uncontrolled fire that spreads rapidly through forests or other vegetation.

Q2. What are the causes of forest fires?

Answer: Dry weather, drought, strong winds and human carelessness are common causes.

Q3. Name some Indian states where forest fires are common according to the chapter.

Answer: Uttarakhand, Himachal Pradesh, Madhya Pradesh and Chhattisgarh.

Q4. What are the effects of forest fires?

Answer: They destroy forests, harm wildlife, damage ecosystems, reduce air quality and displace local communities.

Q5. How can forest fires be prevented?

Answer: People should avoid throwing burning materials into forests, follow fire-safety rules, report fires quickly and support forest protection measures.

## Natural Disasters: Quick Comparison

| Disaster | Main causes | Major effects |
|---|---|---|
| Cyclone | Intense low pressure over the sea and strong winds | Damage to life, property, trees and infrastructure |
| Flood | Heavy rainfall, river overflow or glacial lake burst | Damage to homes, crops, roads and livelihoods |
| Landslide | Heavy rain, earthquakes and unstable slopes | Burial or destruction of houses, roads and farmland |
| Forest fire | Dry conditions, drought, wind and human carelessness | Forest loss, wildlife deaths, poor air quality and economic damage |

# Topic 7: Climate Change

Q1. What is climate change?

Answer: Climate change refers to significant, long-term changes in the climate, including changes in temperature, precipitation and weather patterns.

Q2. What caused climate change in the distant past?

Answer: In the distant past, natural processes caused changes in the Earth's climate.

Q3. What are the main causes of recent climate change?

Answer: Since the 19th century, recent climate change has been largely driven by human activities, especially burning fossil fuels, deforestation, harmful industrial practices and excessive consumption.

Q4. What are fossil fuels?

Answer: Fossil fuels are sources of energy formed from the remains of plants and animals buried deep underground millions of years ago. Coal, petroleum and natural gas are fossil fuels.

Q5. Why does burning fossil fuels affect the climate?

Answer: Burning fossil fuels releases carbon dioxide and other greenhouse gases into the atmosphere. These gases trap extra heat and contribute to global warming.

Q6. What is the greenhouse effect?

Answer: The greenhouse effect is the process by which certain gases in the atmosphere trap heat and keep the Earth warm enough to support life.

Q7. Why is the natural greenhouse effect important?

Answer: It keeps the Earth warm enough for plants, animals and humans to live.

Q8. How do human activities increase the greenhouse effect?

Answer: Human activities release large amounts of greenhouse gases. These gases trap additional heat and cause global warming.

Q9. What is global warming?

Answer: Global warming is the rise in the Earth's average temperature due to the increased trapping of heat in the atmosphere.

Q10. How does climate change affect agriculture?

Answer: It can change rainfall patterns, increase temperatures and disturb crop cycles, reducing agricultural production.

Q11. How can climate change affect small-scale industries?

Answer: Industries that depend on agricultural products, water or predictable weather may face shortages, disruptions and financial losses.

Q12. What is climate change mitigation?

Answer: Mitigation means taking steps to reduce the causes of climate change and slow down global warming.

Q13. Name four measures to reduce climate change.

Answer:

1. Reduce greenhouse gas emissions.
2. Plant and protect trees.
3. Increase the use of renewable energy.
4. Improve energy efficiency and adopt sustainable lifestyles.

Q14. What is resilience?

Answer: Resilience is the ability to withstand difficulties or recover quickly from them.

Q15. What is adaptation to climate change?

Answer: Adaptation means adjusting to actual or expected climate changes and reducing their harmful effects.

Q16. How can communities build resilience to climate disasters?

Answer: Communities can prepare emergency plans, improve drainage, protect forests, follow weather warnings and develop safer infrastructure.

Q17. What is a sustainable lifestyle?

Answer: A sustainable lifestyle uses resources carefully so that they can continue to support people and the environment over a long period.

Q18. What are the consequences of climate change?

Answer: Climate change can increase extreme temperatures, disturb rainfall patterns, affect agriculture and ecosystems, and make life more difficult for people and animals.

Q19. What is the difference between mitigation and adaptation?

Answer: Mitigation reduces the causes of climate change, while adaptation helps people and ecosystems adjust to its effects.

Q20. Why should we protect the environment?

Answer: Protecting the environment helps reduce pollution, conserve natural resources, protect biodiversity and make communities safer and more sustainable.

# Part 8: All Textbook Questions and Activities

The following answers cover the seven numbered questions and activities on pages 65–66 of the uploaded textbook, along with the chapter's important discussion activities.

## Exercise 1: Match the climatic factors with their effects

| Column A | Correct match | Column B |
|---|---|---|
| 1. Latitude | (b) | Creates different climates in the north and south |
| 2. Altitude | (c) | Keeps higher places cooler |
| 3. Proximity to the ocean | (d) | Moderates the temperature |
| 4. Monsoon winds | (a) | Brings wet air to India during summer |

## Exercise 2(a): What is the difference between weather and climate?

Answer: Weather is the condition of the atmosphere at a particular place and time. It changes frequently. Climate is the usual pattern of weather in a region over several decades.

## Exercise 2(b): Why do places near the ocean have milder temperatures than places far away from it?

Answer: The ocean heats up and cools down more slowly than land. It keeps coastal areas cooler in summer and warmer in winter. Therefore, coastal areas have milder temperatures.

## Exercise 2(c): What role do monsoon winds play in affecting India's climate?

Answer: Monsoon winds bring moisture from the ocean and provide rainfall to large parts of India. This rainfall supports agriculture, fills rivers and reservoirs, and influences the lives and livelihoods of millions of people.

## Exercise 2(d): Why is Chennai warm or hot throughout the year, while Leh is cold?

Answer: Chennai is near the sea and has a tropical climate. Its temperatures remain relatively warm throughout the year. Leh is situated at a high altitude in the Himalayas, where temperatures are much lower. Therefore, Leh is much colder than Chennai.

# Exercise 3: Identify the climate of Leh, Chennai, Delhi, Panaji and Jaipur

| City | Climate according to the chapter's categories | Geographical reason |
|---|---|---|
| Leh | Alpine / cold high-altitude climate | It is in the high Himalayan region. |
| Chennai | Tropical climate | It is near the sea and remains warm for much of the year. |
| Delhi | Subtropical climate | It lies in the northern plains and experiences hot summers and cold winters. |
| Panaji | Tropical wet / coastal climate | It is on the western coast and receives heavy monsoon rainfall. |
| Jaipur | Arid to semi-arid conditions | It lies in the dry region of western India, near the Thar Desert. |

Q3(a). Is each place near the sea, in the mountains, or in the desert?

Answer:

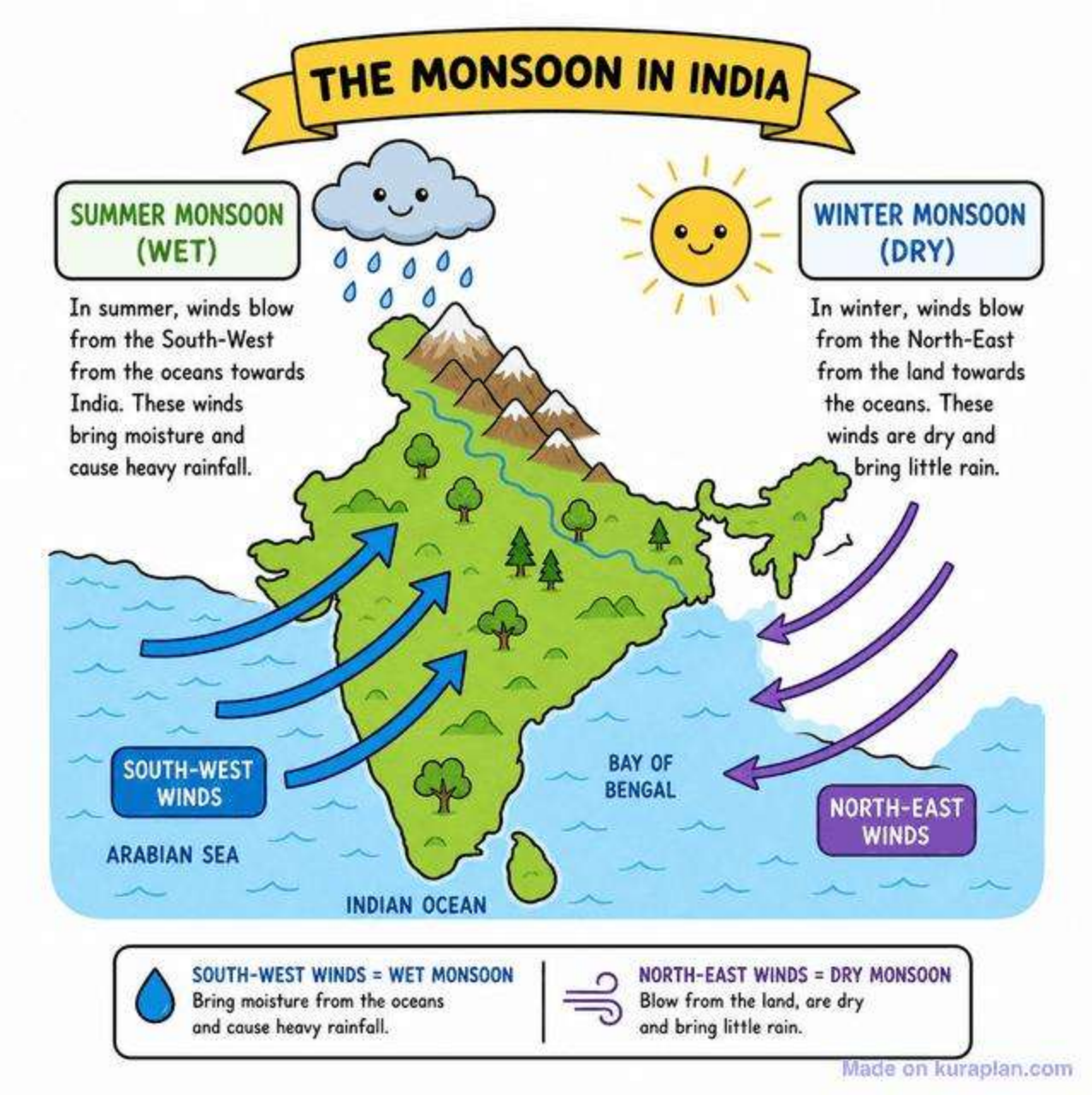
- Leh is in the mountains.
- Chennai is near the sea.
- Delhi is in the northern plains.
- Panaji is near the sea.
- Jaipur is in the dry region near the Thar Desert.

Q3(b). How do these factors affect the climate of these places?

Answer: High altitude makes Leh cold. The sea moderates temperatures in Chennai and Panaji. Delhi's inland location gives it hot summers and cold winters. Jaipur's dry surroundings and limited rainfall contribute to its hot, dry climate.

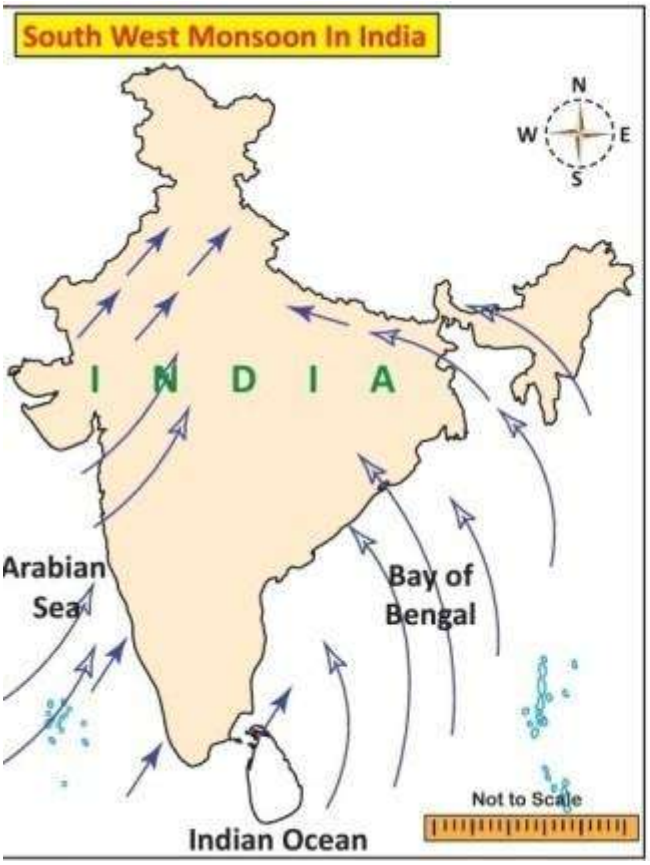
# Exercise 4: Draw the monsoon cycle in summer and winter on a map of India

Answer / Drawing instructions:




A. Summer: Southwest Monsoon

- Draw arrows from the Arabian Sea and Bay of Bengal towards India.
- Show winds advancing from southern India towards the north.
- Label the arrows “Southwest Monsoon”.
- Mark the Western Ghats as a region of heavy rainfall.







B. Winter: Northeast Monsoon

- Draw arrows showing winds blowing from the Indian landmass towards the ocean.
- Label the arrows “Northeast Monsoon”.
- Show some winds crossing the Bay of Bengal and bringing rainfall to parts of eastern and southern India.

# Exercise 5: Make a colourful poster showing festivals linked to farming and weather

Suggested poster content:

# HARVEST & SEAS
# INDIA'S FESTIVA



## BAISAKHI

Harvest festival of Punjab. Makh Nesw & rabi crop harvest. Celebrated with bhangra dance, music, and feats.



## C

Harvest festival of Ke
Features, Pokalam (flo
an



## PONGAL

Harvest festival of Tamil Nada. Sun God (s
(Surya) a is thanked for good harvest,
good harvest. Sweet Pongal dish is cooked.



## L

Winter soltice
Celebrates th
Bonfore, folk song

vedantu.com

# FESTIVALS OF INDIA

### Celebrating Nature, Rain and Harvest

India's festivals connect people with the seasons, farming and nature.

Baisakhi

Harvest festival of Punjab

Pongal

Harvest festival of Tamil Nadu

Onam

Festival of Kerala

Bihu

Festival of Assam

“Save Water • Protect Nature • Respect Farmers”

## Exercise 6: Imagine you are a farmer in India. Write a diary entry about preparing for the rainy season.

**Diary Entry: Preparing for the Rainy Season**

**Date:** 1 June 2026

Dear Diary,

The rainy season is approaching, and I have started preparing my farm. First, I will repair the tools and clean the irrigation channels. I will prepare the soil and select good-quality seeds for sowing.

I will check the weather forecast to learn when the monsoon is expected to arrive. I will also make sure that excess rainwater can drain away from my fields. I will store seeds safely and arrange fertilisers and other necessary materials.

Rainwater is very important for my crops. I hope the monsoon will bring timely and sufficient rainfall. I pray for a good harvest so that my family and I can live happily.

**Good night, Diary!**

## Exercise 7: Write a short essay on a natural disaster, its causes, impacts and preventive measures

### Floods: Causes, Impacts and Prevention

#### Floods: Causes, Impacts and Prevention

A flood occurs when water covers land that is normally dry. Floods are common in many parts of India, especially during the monsoon season.

**Causes:** Floods may occur because of heavy rainfall, overflowing rivers, poor drainage and glacial lake bursts. Unplanned construction and the blocking of natural waterways can increase flood risks.

**Impacts:** Floods can destroy houses, roads, bridges and crops. They may cause loss of life, damage livestock and spread diseases. Many families may lose their homes and livelihoods.

**Preventive Measures:** People should keep drains and waterways clear, avoid dumping waste into rivers and follow official weather warnings. Communities should prepare emergency plans and move to safer places when necessary. The government should improve drainage, maintain flood-protection systems and provide timely warnings and rescue services.

**Conclusion:** Floods can cause serious damage, but proper planning, public awareness and cooperation can reduce their harmful effects.

# Part 9: Additional Important Questions for Examinations

## A. Very Short Answer Questions (1 mark)

Q1. What is the meaning of the Arabic word mausim?

Answer: Season.

Q2. Which mountain ranges protect much of India from cold winds from Central Asia?

Answer: The Himalayas and the Karakoram ranges.

Q3. Which place in India receives about 11,000 mm of average annual rainfall according to the textbook?

Answer: Mawsynram in Meghalaya.

Q4. Which organisation tracks cyclones in India?

Answer: The India Meteorological Department (IMD).

Q5. Which force helps in rescue and evacuation during disasters?

Answer: The National Disaster Response Force (NDRF).

Q6. What is the central region of a cyclone called?

Answer: The eye of the cyclone.

Q7. Name two fossil fuels.

Answer: Coal and petroleum.

Q8. What is the unit used to express relative humidity?

Answer: Percentage (%).

Q9. Which monsoon brings heavy rainfall to India's western coastal strip?

Answer: The southwest monsoon.

Q10. What is the usual period used to describe a region's climate?

Answer: Several decades; the textbook refers to three decades or more.

# B. Short Answer Questions (2–3 marks)

Q1. Explain any three factors that determine India's climate.

Answer:

1. Latitude affects the amount of heat a place receives from the Sun.
2. Altitude makes higher places generally cooler than lower places.
3. Proximity to the sea moderates temperatures in coastal regions.

Winds and topography also play important roles.

Q2. How do mountains affect rainfall?

Answer: Mountains act as barriers to moist winds. The slopes facing the winds may receive heavy rainfall, while areas on the other side may receive less rainfall.

Q3. Explain the relationship between climate and agriculture.

Answer: Climate affects the crops that can grow in a region. Temperature and rainfall influence sowing, irrigation and harvesting. Poor or excessive rainfall can damage crops and reduce farmers' incomes.

Q4. How does climate affect Indian festivals?

Answer: Many Indian festivals are connected with seasons and agricultural activities. Baisakhi, Pongal, Bihu and Onam are examples of festivals associated with harvests and seasonal traditions.

Q5. Explain how urbanisation can increase flood risk.

Answer: Concrete roads and buildings prevent rainwater from soaking into the ground. Poor drainage and construction over natural waterways can cause water to accumulate and flood urban areas.

Q6. How can human activities increase the risk of landslides?

Answer: Cutting forests, unsafe construction and blocking natural drainage can weaken slopes and increase the risk of landslides, especially during heavy rainfall.

Q7. Explain two ways to reduce climate change.

Answer: We can reduce the burning of fossil fuels and increase the use of renewable energy. We can also plant trees, protect forests and use energy more efficiently.

# C. Long Answer Questions (4–5 marks)

Q1. Explain why India's climate is diverse.

Answer: India's climate is diverse because of several geographical factors.

1. Latitude: Different regions receive different amounts of solar heat.
2. Altitude: High mountain regions are colder than the plains.
3. Distance from the sea: Coastal areas have milder temperatures than inland regions.
4. Winds: Winds carry warm or cold air and affect rainfall.
5. Topography: Mountains, plateaus, deserts and valleys influence temperature and precipitation.

These factors together create different climates across India.

Q2. Explain the formation of the summer and winter monsoons.

Answer: In summer, the Asian landmass heats up faster than the ocean. Low pressure develops over the land, and moist winds blow from the relatively high-pressure ocean towards it. These winds bring heavy rainfall to many parts of India. This is the southwest monsoon.

In winter, the land cools faster than the ocean. Pressure becomes relatively higher over the land, and winds blow towards the ocean. These are the northeast monsoon winds. Some of them collect moisture over the Bay of Bengal and bring rainfall to parts of eastern and southern India.

Q3. Explain the effects of climate-related disasters on people and the economy.

Answer: Climate-related disasters can cause loss of life, damage houses and infrastructure, destroy crops and harm animals. Floods can disrupt transport and spread disease. Cyclones can damage coastal settlements, while landslides can destroy roads and buildings. Forest fires destroy vegetation and harm wildlife. These disasters can reduce incomes, interrupt industrial activities and force people to leave their homes. Early warnings, safe construction and disaster preparedness can reduce their impacts.

Q4. What is climate change? Explain its causes and preventive measures.

Answer: Climate change refers to significant, long-term changes in temperature, rainfall and other climate patterns.

Its major recent causes include burning fossil fuels, deforestation, harmful industrial practices and excessive consumption. These activities increase greenhouse gases, which trap additional heat in the atmosphere.

Climate change can be reduced by cutting greenhouse gas emissions, protecting forests, planting trees, using renewable energy and improving energy efficiency. Communities should also prepare for changing weather conditions and natural disasters.

Q5. Explain the relationship between climate, society and the economy.

Answer: Climate affects people's food, clothes, festivals and agricultural practices. Timely monsoon rainfall supports crops and provides water for people and industries. Poor rainfall can cause crop failure, increase food prices and force agricultural workers to migrate. Excessive rainfall can cause floods and damage infrastructure. Therefore, climate plays an important role in society, culture and economic development.

# Final Revision: Learn These Key Facts

| Topic | Key fact to remember |
|---|---|
| Weather and climate | Weather is short-term; climate is a long-term pattern. |

| | |
|---|---|
| Climate factors | Latitude, altitude, proximity to the sea, winds and topography. |
| Summer monsoon | Southwest winds bring moist air from the ocean to India. |
| Winter monsoon | Northeast winds generally blow from land towards the ocean. |
| Mawsynram | About 11,000 mm of average annual rainfall, as stated in the textbook. |
| Cyclones | The IMD tracks cyclones; the NDRF supports disaster response. |
| Floods | Heavy rainfall, river overflow, poor drainage and glacial lake bursts can cause floods. |
| Landslides | Heavy rain, earthquakes and unsafe land use can trigger landslides. |
| Forest fires | Dry conditions, drought, strong winds and human carelessness can cause fires. |
| Climate change | Greenhouse gases trap extra heat and contribute to global warming. |
| Mitigation | Reducing the causes of climate change. |
| Adaptation | Adjusting to climate change and reducing its harmful effects. |

Study tip: First learn the definitions and the five climatic factors. Then practise the monsoon cycle diagram, the comparison tables and the textbook exercises on pages 65–66. Finally, revise the causes, effects and prevention of each natural disaster.